

Summary

- **Data Scientist** with **3+ years** of expertise in data extraction, pre-processing, validation, exploratory data analysis, feature engineering, data wrangling, statistical analysis, machine learning and data visualization
- Expertise in building machine learning models using algorithms such as **Linear Regression, Logistic Regression, Support Vector Machines, Decision trees, KNN, K-means Clustering, and Ensemble methods (AdaBoost, Bagging, Gradient Boosting).**
- Proficient in implementing and fine-tuning deep learning algorithms, including Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), **LSTM, Transformers, BERT, GPT, and Llama.**
- Experience in building production quality and large-scale deployment of applications related to natural language processing (NLP) and machine learning algorithms.

Work Experience

Cigna

Data Scientist

Texas, USA

Jul 2024 – Present

- Established and deployed a **predictive churn model** for healthcare clients, utilizing advanced regression techniques, and identifying critical factors influencing patient attrition, such as treatment outcomes, cost, and access to care, implemented targeted retention strategies, a 20% reduction in patient churn, significantly improving patient satisfaction and loyalty.
- Developed and deployed a recurrent neural network (RNN) using **TensorFlow** for fraud detection in healthcare insurance claims, achieving a 98% accuracy rate in identifying fraudulent claims.
- Built a chatbot using **LangChain** and a pre-trained **LLM** to provide automated customer support for healthcare insurance inquiries, reducing response times by 50% and improving customer satisfaction.
- Implemented predictive models using machine learning algorithms (**random forest, logistic regression**) to forecast patient readmission rates with 70% accuracy, leading to a 50% reduction in healthcare costs.
- Accomplished an **Azure-based NLP pipeline** to analyze customer feedback, leveraging **BERT** embeddings with **Hugging Face Transformers** and **Azure ML** to extract sentiment and key themes, driving a 20% increase in customer satisfaction.

Dixon Technology

Data Scientist

India

Aug 2019 – Jul 2022

- Developed a robust recommendation system utilizing **NLP** techniques and **NLTK library** to enhance product discovery. Extracted meaningful insights from product descriptions and user reviews to create personalized recommendations, resulting in a 20% increase in click-through rates and a 30% boost in sales.
- Achieved a substantial 30% boost in overall model performance by integrating Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN) into a hybrid architecture for complex data analysis tasks
- Created interactive dashboards in **Tableau** to visualize 50+ customer segmentation models, enabling marketing teams to identify high-value target segments and achieve a 20% increase in customer acquisition.
- Trained **XGBoost** model to identify products and promotions correlated with higher customer purchases during promotional seasons. Model insights helped create an optimized promotion strategy for peak season, leading to a 10% increase in demand for promoted products and an 8% growth in overall profits.
- Generated **anomaly detection models** using statistical and machine learning techniques to identify outliers and anomalies in large-scale datasets. This led to a 40% reduction in false positives and improved anomaly detection accuracy.
- Employed a comprehensive suite of data science and machine learning libraries such as **Pandas, NumPy, Seaborn, Matplotlib, Scikit-learn, PyTorch, and NLTK** to refine models and workflows, boosting development speed by 25%.
- Streamlined the **ML model** deployment process by integrating **AWS CodePipeline**, achieving a 50% reduction in deployment time and supporting continuous integration/continuous delivery (CI/CD) for machine learning projects.

Skills

Language: Python, SQL, R
Machine Learning: Linear, Logistic Regression, Decision Trees, Random Forests, Naive Bayes, XGBoost, K-Means
Deep Learning Framework: Tensor-flow, Keras, PyTorch
GenAI: LangChain, LLaMA, Mistral, LlamaIndex, GANs, BERT, GPT-2 and 3
Cloud platforms: AWS (SageMaker, S3, Glue, EMR, Lambda), Azure
Libraries: PySpark, OpenCV, Pillow, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, NLTK, Spacy, Seaborn
Visualizations and Tools: Tableau, Power BI, Git, Bitbucket, Docker, Kubernetes, OpenVino
Database: SQL Server, MySQL, PostgreSQL, MongoDB, Cosmos DB
Operating System: Windows, Linux, IOS

Education

Master of Science in Business Analytics

Trine University – Indiana, USA

May 2024